

MILESTONE 1 PIPELINE

Day 3 – Development Begins

Core Pipeline Execution: Repository Upload → Analysis → AI Response → Architecture JSON

Team Assignments & Tasks

Team 1 – Frontend Team

Sanchita, Pragya

TASKS

- UI Design specification mapping
- Set up the Next.js frontend architecture
- Create core views:
 - Landing Page
 - Repository Upload Page
 - Loading State Screen
 - Architecture Viewer Page (placeholder)

DELIVERABLE

A fully working frontend prototype with seamless navigation implemented between all specified pages.

Team 2 – Backend Team

Rajvir, Srijoyee

TASKS

- Establish the Express backend project structure
- Implement required API routes:
 - POST /upload — Handle incoming repository files
 - POST /analyze — Trigger processing structure
 - GET /health — Service health check endpoint
- Build the repository scanner utility to parse directories
- Format and return the scanned repository structure as a JSON tree

DELIVERABLE

Successful upload of a sample repository returning a structured JSON file tree representation.

Team 3 – AI Team

Vaibhav, Oitrika

TASKS

- Finalize systemic Gemini prompts for accurate parsing
- Create the dedicated Gemini integration service layer
- Ingest the repository JSON tree structured by the backend
- Generate structured architecture JSON output containing nodes and edges

TARGET OUTPUT FORMAT

```
{
  "nodes": [],
  "edges": []
}
```

DELIVERABLE

End-to-end prompt workflow mapping: Repository Structure JSON → Gemini Engine → Validated Architecture JSON.

Integration Goal

By the end of Day 3, the target objective is a complete functional demonstration of the end-to-end flow. Even with a basic interface, the data pipeline must resolve sequentially:



HARD DEADLINE

Tomorrow, 8:00 PM IST